



Electrical Resistivity Measured by Millisecond Pulse Heating in Comparison with Thermal Conductivity of the Superalloy Inconel 625 at Elevated Temperature

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Abstract

Selected thermophysical properties of Inconel 625 were measured in continuation of our work on the comparison between thermal conductivity and electrical resistivity for different alloys. In contrast to pure metals, alloys usually show significant deviations from the Wiedemann–Franz law using the theoretical Sommerfeld value. Two experimentally determined constants can take into account mainly lattice and electron scattering contributions (Smith–Palmer plot), re-establishing a well-defined relation between thermal and electrical conductivity. Thermal diffusivity of Inconel 625 was measured by the laser flash method in the temperature range –120 °C to 1250 °C; heat capacity was measured by differential scanning calorimetry in the temperature range –170 °C to 1250 °C; thermal expansion was measured by dilatometry in the temperature range –150 °C to 1295 °C (solidus temperature). Density at room temperature was measured by an Archimedeian balance. From these experimentally obtained data, thermal conductivity was calculated in a wide temperature range. Electrical resistivity of Inconel 625 was measured by millisecond pulse heating in the temperature range from room temperature to the solidus temperature. The measurement results of electrical resistivity as a function of specific enthalpy were combined with results of specific heat capacity measurements to obtain the relation between resistivity and temperature.

Keywords Electrical resistivity · Inconel 625 · Millisecond pulse heating · Smith–Palmer plot · Specific heat capacity · Thermal conductivity

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1 Introduction

The superalloy Inconel 625 (steel number 2.4856) shows excellent corrosion resistance in a wide range of temperatures and pressures and also high ductility. Originally designed for applications in steam and gas turbines, this alloy is widely used in seawater and chemical processing. Two different grades of the alloy are established for different service temperatures. Grade 1 is described as “soft annealed,” annealed between 930 °C and 1035 °C, with a fine grain structure and good mechanical and exceptional corrosion resistance properties for service temperatures up to 600 °C. Grade 2 is described as “solution annealed,” annealed between 1080 °C and 1160 °C, with a coarser grain structure and excellent creep strength in high-temperature applications above 600 °C [1–4].

There is very little thermophysical property data of this material in the low and high temperature range reported in the publicly available literature [5–7], although accurate data are urgently needed for the numerical simulation of industrial processes like casting, laser cladding or additive manufacturing.

In this work, the measurements of thermophysical properties (electrical resistivity, thermal diffusivity and conductivity, as well as heat capacity) of the superalloy Inconel 625 in the temperature range from below room temperature very close to the solidus temperature of 1295 °C are reported. The relation between thermal and electrical conductivities is calculated using a Smith–Palmer plot and compared to the prediction of the Wiedemann–Franz law.

2 Measurements

2.1 Specimen Material

The superalloy Inconel 625 was supplied by voestalpine BÖHLER Edelstahl GmbH & Co KG, Kapfenberg, Austria. The specimens were cut from bulk material, which was heat-treated at 930 °C in nitrogen for 1 h. The chemical composition is given in Table 1. Carbon and sulfur were determined by combustion and detection by infrared spectroscopy (apparatus: LECO CS744, Leco Inc., Saint Joseph, MI, U.S.A.); silicon was determined by a gravimetric method, and all other elements were determined by optical ICP emission spectroscopy (apparatus: Vista-MPX, Varian, Palo Alto, CA, U.S.A.) after acid hydrolysis. Nine individual specimens were machined for the measurement of electrical resistivity, six individual specimens for all other measurements.

Table 1 Chemical composition of the investigated superalloy Inconel 625 in percent by mass

Elements												
C	Cr	Mo	Co	Nb + Ta	Al	Ti	Fe	Mn	Si	P	S	Ni
0.022	20.2	8.0	0.34	4.0	0.46	0.16	3.4	0.10	0.28	0.006	<0.002	Bal.

2.2 Specific Heat Capacity

The heat capacity of the specimens was measured by two individual heat flow differential scanning calorimeters of NETZSCH-Gerätebau, Selb, Germany, type DSC 404 C/3/G Pegasus and type DSC 404 F1 Pegasus. The measurements were performed as described in [8] and according to EN 821-3 (2005) [9]. Sapphire and molybdenum SRM 781 (National Institute of Standards and Technology, Gaithersburg, MD, USA) with certified heat capacity values were used as reference materials [10] for the heat flow rate calibration. The individual mass of the specimens (thickness, 1.5 mm; diameter, 6 mm) was measured by an analytical balance. The temperature calibration of the DSC instruments was performed by determining onset temperatures of phase changes of pure materials (adamantane, indium, zinc, aluminum and gold) to cover most of the temperature range. In the low temperature range, uncertainty in temperature measurement is contributing more significant to the uncertainty of heat capacity. In the very high temperature range, heat capacity is only slightly changing with temperature. The crucible material was platinum (with alumina inserts at temperatures above 1100 °C); the measurements were made at a constant heating/cooling rate of 10 K·min⁻¹ in an argon (99.999 %) environment.

2.3 Thermal Expansion and Density

The thermal expansion of the superalloy Inconel 625 was measured by dilatometry according to DIN 51045-1 (2005) [11], and measurement details and results are reported elsewhere [12]. The density of the specimens at room temperature was determined by an Archimedean balance (Sartorius, type LA 230S-OCE). The temperature dependence of the density (specific mass), ρ , was calculated from the density at room temperature, ρ_0 , and the linear thermal expansion, $\Delta l/l_0$, by

$$\rho(T) = \rho_0 \frac{1}{\left(1 + \frac{\Delta l(T)}{l_0}\right)^3}. \quad (1)$$

2.4 Thermal Diffusivity

The thermal diffusivity of the superalloy Inconel 625 was measured by two commercial laser flash apparatus (NETZSCH-Gerätebau, both type LFA 427) as described in [8] and according to EN 821-2 (1997) [13]. The thickness (nominally 3 mm) and diameter (nominally 12.55 mm) of the specimens were measured by a calibrated precision caliper. In order to improve the optical properties, the specimens were carefully roughened by slight sandblasting. The measurement was performed at low temperature in helium (99.999 %) and at elevated temperatures in high vacuum (10⁻⁵ mbar). The measured value of thermal diffusivity, a_0 , is related to room temperature dimensions of the specimen and therefore corrected for thermal expansion, $\Delta l/l_0$, at temperature, T , to obtain the true thermal diffusivity, a , as:

$$a(T) = a_0 \left(1 + \frac{\Delta l(T)}{l_0} \right)^2. \quad (2)$$

The correction was done with the measured values given in [12].

2.5 Electrical Resistivity

The electrical resistivity of the superalloy Inconel 625 was measured by a four-point probe using a millisecond pulse-heating system and applying heavy-duty batteries to supply a current of up to 4000 A. The current through the cylindrical specimen is determined by measuring the voltage across a standard resistor connected in series with the specimen. The voltage across the middle part of the specimen is measured between spring-loaded knife-edge probes. They are mounted on stationary clamps via a lever mechanism that allows the probes to move in the axial direction to follow the thermal expansion of the specimen during the experiment. Current through the specimen and voltage along a defined portion of the specimen are recorded simultaneously every 0.5 ms by a data acquisition system with sample-and-hold amplifiers with full-scale resolution of 16 bit. The heating rate during measurement was approximately $5000 \text{ K}\cdot\text{s}^{-1}$. A detailed description of the construction and operation of the pulse-heating system is given in earlier publications [14–17]. The specimens were cylindrical with a nominal length of 75 mm (40 mm between the voltage probes) and 4 mm in diameter.

The measurement principle (transient four-point probe) requires a long, thin rod to avoid (i) axial heat flow to the clamps, and (ii) radial heat flow due to cooling of the surface by radiation. Numerical finite element simulations were performed to estimate the additional uncertainties in the measured electrical resistivity due to the finite length and the relatively large diameter. With the worst-case scenario (fixed room temperature at the clamp positions, total hemispherical emissivity of 100 %), the simulations showed that the heat flow to the clamps is negligible; the heat flow by radiation to the surrounding increased the measurement uncertainty of electrical resistivity by approx. 0.1 % at the highest temperatures.

The electrical resistivity, ρ_{el} (corrected for thermal expansion of the room temperature diameter, d_0 , and length between the voltage probes, l_0 , by thermal expansion, $\Delta l/l_0$), as a function of time, t , is calculated from the voltage drop, U , and the current, I , by

$$\rho_{el}(t) = \frac{U(t)d_0^2\pi}{4I(t)l_0} \left(1 + \frac{\Delta l(T)}{l_0} \right). \quad (3)$$

The specific enthalpy, H , as a function of time, t , is calculated by the voltage, U , current, I , and mass between the knife edges, m , as:

$$H(t) = \frac{1}{m} \int_0^t U(\tau)I(\tau)d\tau. \quad (4)$$

In order to obtain the relation between the enthalpy and temperature, the results of the previously measured specific heat capacity (given in Sect. 3.1) were integrated over temperature. With these results, the electrical resistivity, enthalpy and temperature were known at each time step during the pulse heating experiment. By eliminating the time basis, the mutual dependency between resistivity and temperature was obtained.

3 Results

3.1 Specific Heat Capacity and Thermal Diffusivity

The results of the measurements of the specific heat capacity and the thermal diffusivity of the superalloy Inconel 625 are shown in Tables 2 and 3, as well as in Fig. 1 in comparison with results of other authors [5–7]. The expanded uncertainty for the specific heat capacity measurement is estimated to be $\pm 3\%$ in the low and medium temperature range and up to $\pm 5\%$ above 1100 °C. All uncertainties of measurements reported in Sect. 3 of this work are stated as the standard uncertainty

Table 2 Low-temperature thermophysical data of superalloy Inconel 625 as a function of temperature (specific heat capacity, density, thermal diffusivity corrected for thermal expansion, calculated thermal conductivity)

Temperature (°C)	Heat capacity ($\text{J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$)	Density ($\text{kg}\cdot\text{m}^{-3}$)	Thermal diffusivity ($\text{mm}^2\cdot\text{s}^{-1}$)	Thermal conductivity ($\text{W}\cdot\text{K}^{-1}\cdot\text{m}^{-1}$)
–170	0.244			
–160	0.266			
–150	0.286	8501		
–140	0.303	8499		
–130	0.318	8497		
–120	0.331	8494	2.62	7.37
–110	0.343	8491	2.62	7.62
–100	0.353	8489	2.62	7.85
–90	0.362	8486	2.63	8.07
–80	0.370	8483	2.63	8.27
–70	0.378	8480	2.64	8.46
–60	0.384	8477	2.66	8.64
–50	0.390	8474	2.67	8.83
–40	0.395	8471	2.69	9.01
–30	0.400	8468	2.71	9.19
–20	0.405	8465	2.74	9.37
–10	0.409	8462	2.77	9.56
0	0.412	8459	2.80	9.75
10	0.416	8456	2.83	9.95

Table 3 High-temperature thermophysical data of superalloy Inconel 625 as a function of temperature (specific heat capacity, density, thermal diffusivity corrected for thermal expansion, calculated thermal conductivity, electrical resistivity corrected for thermal expansion)

Temperature (°C)	Heat capacity (J·g ⁻¹ ·K ⁻¹)	Density (kg·m ⁻³)	Thermal diffusivity (mm ² ·s ⁻¹)	Thermal conductivity (W·K ⁻¹ ·m ⁻¹)	Electrical resistivity (μΩ·m)
20	0.419	8453	2.86	10.1	1.265
50	0.427	8444	2.97	10.7	1.268
100	0.439	8429	3.13	11.6	1.274
150	0.448	8413	3.30	12.4	1.280
200	0.457	8397	3.47	13.3	1.285
250	0.465	8380	3.63	14.1	1.290
300	0.472	8363	3.79	15.0	1.296
350	0.480	8345	3.96	15.8	1.301
400	0.487	8326	4.12	16.7	1.306
450	0.494	8307	4.28	17.6	1.311
500	0.501	8288	4.44	18.4	1.315
540	0.507	8272	4.56	19.1	1.319
550	0.518	8268	4.59	19.4 ^a	1.320
600	0.575	8247	4.75	20.3 ^a	1.324
610	0.586	8243	4.78	20.5 ^a	1.325
650	0.589	8226	4.91	21.2 ^a	1.329
700	0.594	8204	5.06	22.0 ^a	1.333
720	0.596	8195	5.12	22.4 ^a	1.334
740	0.598	8187	4.99	22.7 ^a	1.336
750	0.599	8182	4.92	22.9 ^a	1.333
770	0.601	8173	4.79	23.5	1.327
800	0.604	8160	4.86	24.0	1.318
840	0.608	8141	4.97	24.6	1.306
850	0.609	8137	4.99	24.7	1.306
900	0.614	8113	5.12	25.5	1.304
950	0.620	8089	5.25	26.3	1.303
1000	0.625	8065	5.39	27.2	1.302
1050	0.631	8040	5.52	28.0	1.303
1100	0.638	8014	5.65	28.9	1.304
1150	0.644	7988	5.78	29.8	1.305
1200	0.651	7962	5.92	30.7	1.307
1250	0.657	7935	6.05	31.6	1.310
1290	0.663 ^a	7914	6.16 ^a	32.3 ^a	1.313

^aInter/extrapolated

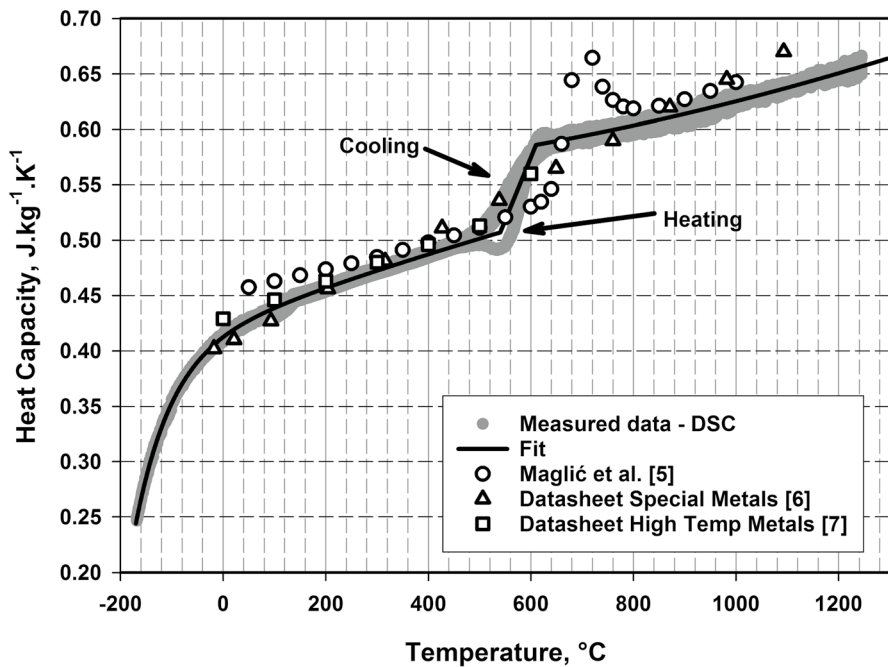


Fig. 1 Heat capacity of the superalloy Inconel 625 as a function of temperature in comparison with results of other authors and datasheet information

of measurement multiplied by the coverage factor $k=2$, which for a normal distribution corresponds to a coverage probability of approximately 95 %. A kink can be noticed in the heat capacity curve between 500 °C and 620 °C during heating and cooling, which is in the temperature range where forming and dissolution of precipitations are expected. The uncertainty for the thermal diffusivity measurement is estimated to be ± 4 % in the temperature range up to 1000 °C, and ± 5 % at higher temperatures.

3.2 Thermal Conductivity

In temperature ranges where no thermal reactions occur, the thermal conductivity, λ , as a function of temperature can be calculated from thermal diffusivity related to room temperature dimensions of the specimen, a_0 , density at room temperature, ρ_0 , specific heat capacity, c_p , and linear thermal expansion, $\Delta l/l_0$, by the following equation:

$$\lambda(T) = \frac{a_0(T)\rho_0 c_p(T)}{1 + \frac{\Delta l(T)}{l_0}}. \quad (5)$$

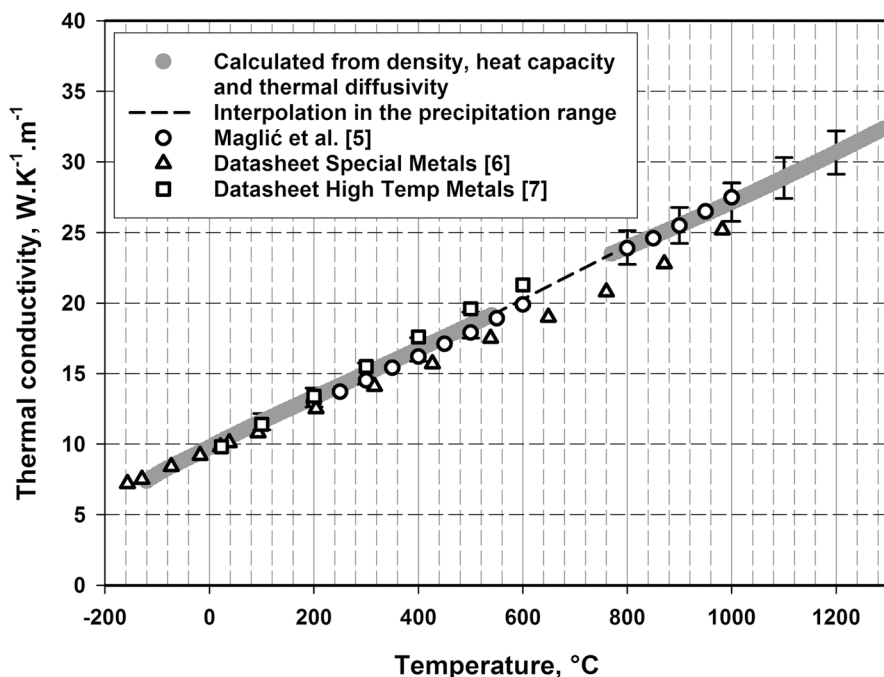


Fig. 2 Thermal conductivity of the superalloy Inconel 625 as a function of temperature in comparison with results of other authors and datasheet information

The results of the obtained values for the thermal conductivity of the superalloy Inconel 625 are shown in Tables 2 and 3, as well as in Fig. 2 in comparison with results of other authors [5–7]. The temperature region 550 °C to 760 °C, where thermal reactions due to precipitation forming/dissolving occur, is interpolated by a linear function. The uncertainty for thermal conductivity is estimated to be $\pm 5\%$ in the temperature up to 1000 °C, and $\pm 6\%$ at temperatures higher than 1000 °C.

3.3 Electrical Resistivity

The electrical resistivity data were first computed with room temperature specimen dimensions and in a second step—with the dilatometer data—corrected for thermal expansion. All reported data of electrical resistivity are corrected for thermal expansion and relate to the true geometry of the specimen at high temperature. The variation of electrical resistivity as a function of temperature of the superalloy Inconel 625 is shown in Table 3 and in Fig. 3 in comparison with results of other authors [5–7]. The repeatability of the measurements (4 repetitions) for an individual specimen is 0.3 %, and that between different specimens is 0.6 %. The expanded uncertainty of electrical resistivity is estimated to be $\pm 1.7\%$.

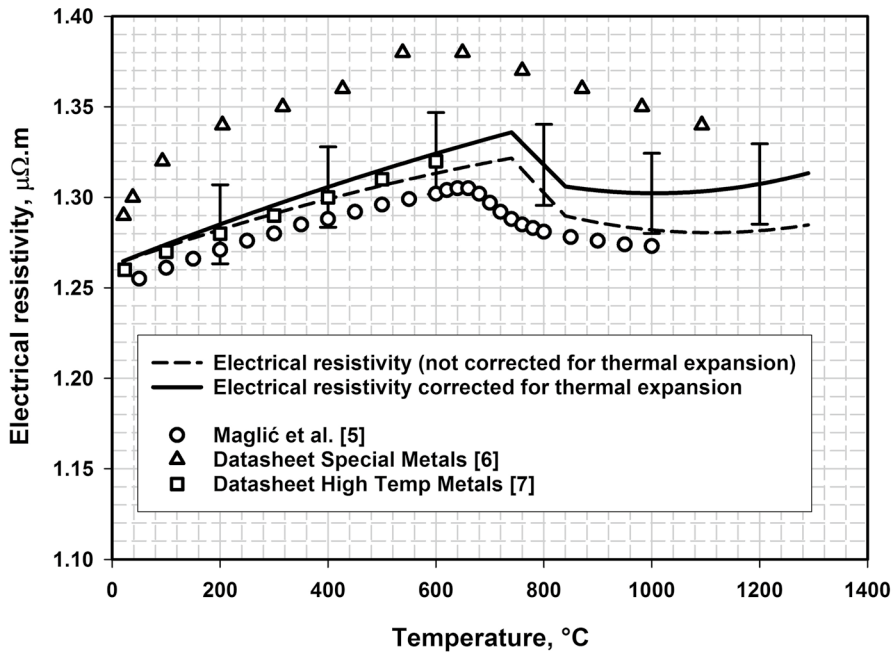


Fig. 3 Electrical resistivity of the superalloy Inconel 625 as a function of temperature in comparison with results of other authors and datasheet information (results of Maglič et al. are not corrected for thermal expansion, the datasheets give no information)

4 Comparison Between Thermal Conductivity and Electrical Resistivity

In well conducting metals and alloys, the contributions to thermal conductivity are mainly electronic and only minor contributions arise from the lattice component [18]. Theoretically, at moderate temperatures, T , the electronic component, λ_{el} , approximately obeys the Wiedemann–Franz–Lorenz relation

$$\lambda_{el}(T) = \frac{L_0 T}{\rho_{el}(T)}, \quad (6)$$

where L_0 is the Sommerfeld value of the Lorenz number ($2.445 \times 10^{-8} \text{ V}^2 \cdot \text{K}^{-2}$), ρ_{el} is the electrical resistivity. This equation was modified by Smith and Palmer [19] to account for the scattering of electrons by impurities and for the lattice component of thermal conductivity

$$\lambda(T) = \frac{AL_0 T}{\rho_{el}(T)} + B, \quad (7)$$

with A and B as empirical constants.

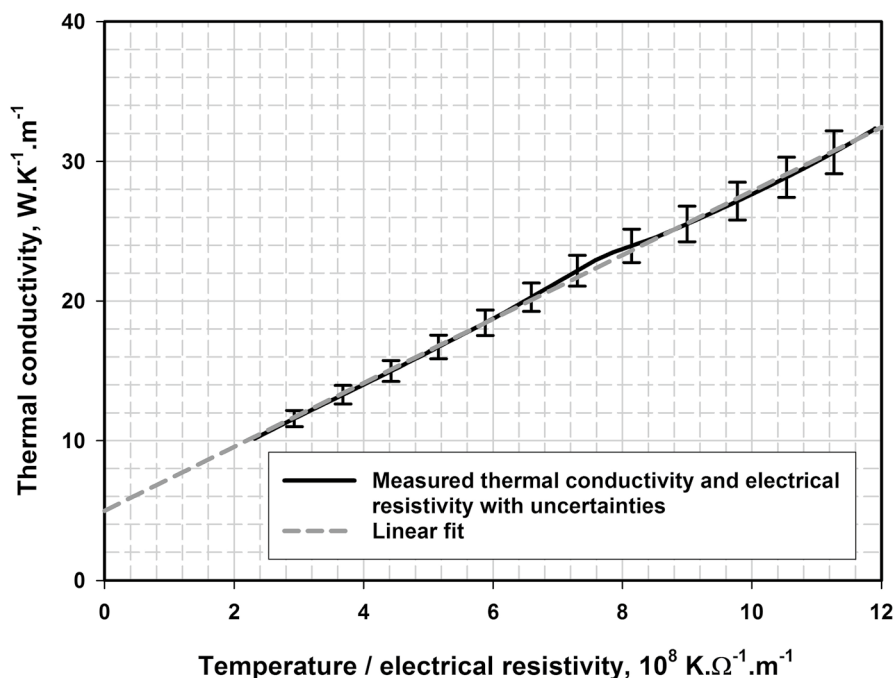


Fig. 4 Thermal conductivity of the superalloy Inconel 625 as a function of temperature over electrical resistivity (black line: calculated from thermal diffusivity, heat capacity and density measurements; gray dashed line: linear fit to the measured values between $2.3 \times 10^8 \text{ K} \cdot \Omega^{-1} \cdot \text{m}^{-1}$ and $11.9 \times 10^8 \text{ K} \cdot \Omega^{-1} \cdot \text{m}^{-1}$)

Figure 4 shows the results of thermal conductivity (calculated from thermal diffusivity, heat capacity and density) as a function of temperature over electrical resistivity including the uncertainty bars. A linear equation was fit to the measured values between $2.3 \times 10^8 \text{ K} \cdot \Omega^{-1} \cdot \text{m}^{-1}$ and $11.9 \times 10^8 \text{ K} \cdot \Omega^{-1} \cdot \text{m}^{-1}$ (corresponding to values from room temperature to the solidus temperature at 1295°C). The best fit to the data was obtained for the constants A and B , 0.94 and $5.0 \text{ W} \cdot \text{K}^{-1} \cdot \text{m}^{-1}$, respectively.

Figure 5 finally shows the results of the thermal conductivity as a function of temperature obtained by measurements of thermal diffusivity and electrical resistivity using the Smith–Palmer plot, as well as the noticeable deviating results obtained by the Wiedemann–Franz law.

5 Discussion

Considering the wealth of the literature data on the superalloy Inconel 718, there is remarkable little information on the thermophysical properties of the superalloy Inconel 625. The only work on thermophysical properties of Inconel 625

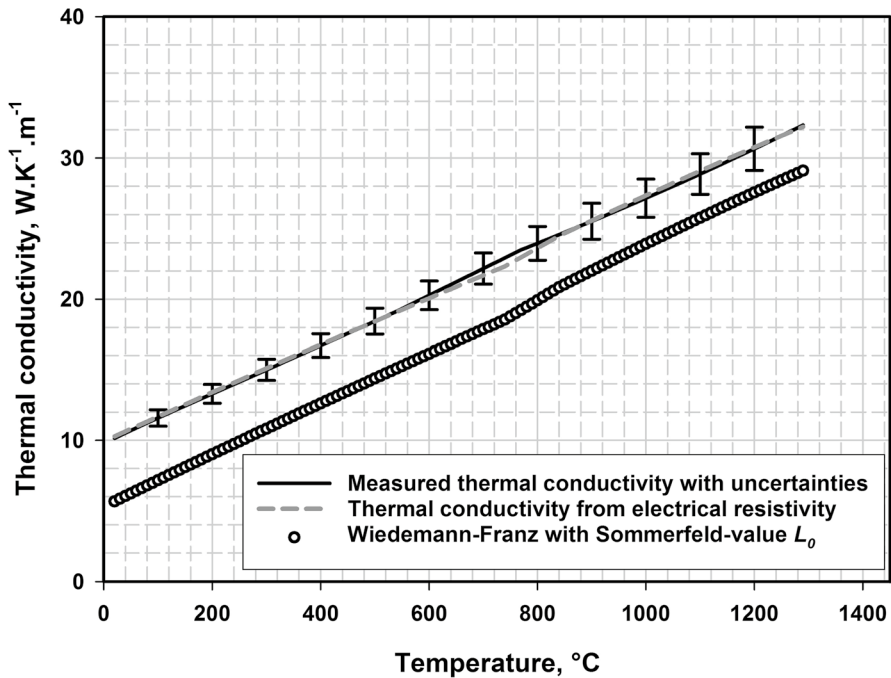


Fig. 5 Thermal conductivity of the superalloy Inconel 625 as a function of temperature (black line: measurements; gray dashed line: calculated from electrical resistivity using a Smith–Palmer plot with constants $A=0.94$ and $B=5.0$ in Eq. 7; dots: Wiedemann–Franz law)

published in a scientific journal has been reported by Maglič et al. [5], specifying also measurement uncertainties [20]. Two other sources (datasheets) from the internet [6, 7] show thermophysical data, but they neither report information on measurement conditions nor show information on measurement quality. However, since no other information is available, these data are included in the discussion.

Maglič et al. [5] report specific heat of Inconel 625 in the temperature range 50 °C to 1000 °C. The values agree well with those of the present work, except in the lower temperature range and in the range of the formation/dissolution of the precipitates. At 50 °C, the heat capacity is 7 % higher, which is slightly higher than the combined uncertainties. Maglič et al. noted that the relatively high heating rate applied might be the cause for a retarded dissolution of the precipitates at approx. 700 °C. There is a quite good agreement between the datasheets from the internet [6, 7] and the present work.

Maglič et al. [5] report thermal conductivity of Inconel 625 in the temperature range 200 °C to 1000 °C which is in a remarkable good agreement with the present work. These data are not corrected for thermal expansion, which is 0.66 % at 500 °C and 1.61 % at 1000 °C [12]. Considering the correction, the results of Maglič et al. are slightly lower but still in excellent agreement with this work. The maximum deviation is 3 % at 300 °C. The datasheet of one of the producers

of Inconel 625 [6] presents noticeable lower results; another producer [7] shows somewhat higher values for thermal conductivity.

The reported values for electrical resistivity of Inconel 625 of Maglić et al. [5] in the temperature range 50 °C to 1000 °C are lower than the present work, but agree within the combined uncertainties. One of the internet sources [6] presents significantly higher results, the second one [7] is in almost perfect agreement.

Experimentally observed constants for slope A and intercept B in Eq. 7 for the superalloy Inconel 625 are not available in the literature. Klemens [18] lists parameter for a variety of similar stainless steels and nickel-based superalloys; typical values of constant A range from 0.81 to 0.98, which means electron scattering by impurities lowers thermal conductivity by approx. 2 % to 19 %, while typical values of constant B range from $4.5 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ to $7.4 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which accounts mainly for the additional heat transfer by the lattice. Values reported for stainless steel AISI 316L [17] are for A and B , 0.94 and 5.0, respectively. This is a noticeable agreement with the results of the present work.

6 Conclusions

The following conclusions can be drawn from the present work:

- An extended set of thermophysical properties from below room temperature to melting was measured to characterize the superalloy Inconel 625, including electrical resistivity, specific heat capacity, density at room temperature and thermal diffusivity. From these results, thermal conductivity and density as a function of temperature were calculated.
- A modified Wiedemann–Franz law has to be used to calculate thermal conductivity from electrical resistivity—the modification is essential as the lattice contribution for the superalloy Inconel 625 is relatively high.
- Electrical measurements can be used to estimate thermal conductivity for this type of superalloy quite accurately.
- Electrical resistivity can be measured usually much more precisely (e.g., millisecond pulse heating) than thermal conductivity at very high temperature: therefore, the calculation of thermal conductivity by a Smith–Palmer equation from electrical measurements is an interesting alternative to thermal conductivity (or diffusivity) measurements.

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